

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	A passing photon with energy equal to 1.79 eV (or equivalent) (1) Causes another photon {of same frequency / same energy / in phase} to be emitted (1) accept similar photon emitted	2			2		
		(ii)	So stimulated emission is more {frequent / probable / likely} than absorption	1			1		
	(b)		$\lambda = \frac{hc}{\Delta E}$ or $f = \frac{\Delta E}{h}$ and $\lambda = \frac{c}{f}$ transposition at any stage (1) $\Delta E = 2.86 \times 10^{-19}$ J or by implication (1) $\lambda = 694$ n[m] (1) ecf on ΔE		3		3	2	
	(c)	(i)	Number per second = $\frac{0.60}{2.86 \times 10^{-19} \text{ecf}}$ or $2.1 \times 10^{18} \text{ [s}^{-1}\text{]}$		1		1	1	
		(ii)	$p = \frac{6.63 \times 10^{-34}}{6.94 \times 10^{-7} \text{ecf}}$ or by implication ecf (1) $p = 9.6 \times 10^{-28}$ N s or kg m s ⁻¹ unit mark (1)		2		2	2	
		(iii)	Momentum per second = Answer to (i) $\times$ Answer to (ii) or 2.0×10^{-9} [N] (1) Force on reflecting surface = $2 \times 2.0 \times 10^{-9} = 4.0 \times 10^{-9}$ [N] ecf (1) Award 1 mark only for multiplication by 2 anywhere		2		2	2	
			Question 9 total	3	8	0	11	7	0